Zhouyuan Chen

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Education

Sep. 2023 – May 2025	New York University, New York, NY, United States Master of Science(M.Sc.), Computer Science GPA:3.778/4
Sep. 2019 – Jun. 2023	Zhejiang University of Technology, Hangzhou, China Bachelor of Engineering(B.E.), Software Engineering GPA:88.6/100

Research Interests

Computer Graphics, Geometry Processing, Numerical Simulation, Shape analysis, Digital fabrication

Publications & Preprints

1. **Zhouyuan Chen**, Riso, M., Baas, B., Schulz, A. & Bousseau, A. Differentiable Stock Constraints for Continuous Optimization of Frame Structures by Reuse. *ACM Symposium on Computational Fabrication (SCF), 2026.*
2. Zint, D., **Zhouyuan Chen**, Zhu, Y., Zorin, D., Schneider, T. & Panozzo, D. Topological Offsets. *ACM Transaction on Graphics (SIGGRAPH), 2025.*
3. **Zhouyuan Chen**, Kusupati, U., Liu, Y., Zint, D., Zorin, D., Panozzo, D. & Schneider, T. Prismatic Mesh Generation for Volumetric Rods and Shells. *Manuscript under revision for resubmission.*
4. Zint, D., Kusupati, U., Bertram, S., **Zhouyuan Chen**, Schneider, T., Zorin, D. & Panozzo, D. SimWild: Physical Simulation on Imperfect Geometry in the Wild. *Manuscript under revision for resubmission.*

Experience

Aug. 2026 – Present	Independent Researcher Collaborators: Jiong Chen and Teseo Schneider • Developing a 2D/3D cage deformation framework based on micropolar elasticity, introducing boundary microrotation as an additional deformation control.
Mar. 2026 – Jul. 2026	Inria Sophia Antipolis, Nice, France Research Intern at GraphDeco Team Advisor: Adrien Bousseau • Designed a novel differentiable topology optimization framework for frame structure design.
Sep. 2025 – Dec. 2025	Inria Nancy – Grand Est, Nancy, France Research Engineer at Pixel Team Advisor: Etienne Corman • Studied and implemented several conformal parameterization algorithms and medial axis methods, and developed an algorithm for computing the medial axis on curved surfaces.
Jun. 2025 – Jan. 2026	University of Victoria, Victoria, BC, Canada Research Assistant (Remote) Advisor: Teseo Schneider • Developed a robust algorithm for constructing prism-dominant shell meshes for shell-and-rod numerical simulations.
Jul. 2023 – May 2025	New York University, New York, NY, United States Research Assistant at Geometric Computing Lab Advisor: Daniele Panozzo

• Designed a pipeline to automate biomechanical simulations and implemented a plugin to streamline workflows.
• Designed and implemented an offsets generation algorithm and contributed to the Wildmeshing Toolkit C++ library.

Sep. 2021 – Jun. 2023 Zhejiang University of Technology, Hangzhou, China
Research Assistant at Institute of Digital Media Technology
Advisor: Jiazhou Chen

• Implemented several broad-phase collision detection algorithms and integrated them into medical software while also developing a custom algorithm to visualize collisions between different objects.
• Designed and implemented a custom algorithm to reconstruct the undercut model of human teeth.

Teaching Experience

New York University

Teaching Assistant
Geometric Modeling(CSCI-GA.3033-018) Spring 2024, Spring 2025

Grader
Intro to Computer Science(CSCI-UA.101) Spring 2024

Technical Skills

Programming Language: C++, C, Python, Java, MATLAB
Languages: Chinese (native), English (fluent)

References

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Etienne Corman
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